

# Project Status

## Project Status — Complete Explanation

*Last updated: September 15, 2026*

This is the single comprehensive document explaining the whole project: what the assignment is, what's been done, what the current bull/base/bear numbers and final call are, what the reference deck says and how we're using it, what tables/charts the eventual presentation will contain, and an honest review of whether the work actually satisfies the brief. Read this first if you're picking the project back up cold.

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### 1. The assignment, restated

Jefferies (Healthcare Equity Research) sent a take-home test: produce a PowerPoint “deep dive” on Amgen’s dazodalibep Phase 3 program in Sjögren’s disease (**NCT06104124** and **NCT06245408**) and land on an explicit, evidence-backed call — will the primary efficacy endpoint(s) read out positive or negative. Required topics (not exhaustive): biological rationale of CD40L targeting + Phase 2 insights; unique features of the Phase 3 designs (symptomatic vs. systemic) + statistical considerations; lessons from other trials (ianalumab). A reference deck (MDGL/ MAESTRO-NASH) was supplied as a template for analytical depth and slide anatomy, not as the topic. Due Friday, September 18, 11:00am ET.

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### 2. What’s been done — the process, end to end

#### Day 1 (foundational research) — complete

Pulled and verified: the Phase 2 published paper (both patient populations), both Phase 3 trials’ official registrations and current status, the ESSDAI measurement literature (MCID, placebo-response modeling, composite indices), the ianalumab competitor program (Phase 2b and Phase 3), a historical table of comparator Sjögren’s trials, and a risk note for the secondary trial. See `README.md` for the file-by-file index.

#### Day 2 (quantitative modeling) — complete

Built a statistical bridge from Phase 2’s observed result to a Phase 3 forecast: a minimum-detectable-effect / variance-sensitivity analysis, a bull/base/bear scenario table, and an explicit final thesis with a ranked counter-argument.

#### External review and correction pass — complete

An outside review (`Jefferies_Project_Review.md`) caught several real errors — most importantly, that ianalumab’s Phase 2b trial was mischaracterized as a “miss” (it actually met its real primary objective, a dose-response test), that a statistical calculation understated variance and overstated confidence, and that the secondary trial’s second co-primary endpoint (DASPRI) was missing from the narrative entirely. All of these were independently re-verified against primary sources (not accepted at face value — the review itself had some errors too, documented in `phase2_baseline_characteristics.md`) and corrected across every affected file.

#### Additional topics pass — partially complete

Of 8 “additional topics worth adding” the review suggested, 3 were judged high-value/ low-effort and completed (dose/exposure PK-PD bridge, clinical-meaning synthesis, readout interpretation table — see `day2_additional_topics.md`); 2 were already covered as a byproduct of the correction pass (NEPTUNUS-1/2 individual results, iscalimab’s symptomatic-cohort miss); 3 were deprioritized as low-payoff for a 4-day assignment (missing-data/rescue-medication treatment, and deeper endpoint-architecture/ population-transportability work beyond what’s already captured).

## Not yet started

Day 3 (slide-by-slide text outline) and Day 4 (actual .pptx build + PDF export).

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## 3. The current bull/base/bear numbers and final call

**This is the load-bearing quantitative result of the whole project.** Full detail and methodology in `day2_scenario_model.md`; the call itself in `day2_final_thesis.md`.

### The statistical setup

NCT06104124 has N=651 across 3 arms (2 dazodalibep doses + placebo; ~217/arm assumed, since the actual allocation ratio isn’t disclosed). Three different variance (standard deviation) assumptions were tested, because picking just one — as an earlier version of this project did — silently determines how confident the whole forecast looks:

| Variance assumption | Source                                                                                                   | Effect needed for 80% power |
|---------------------|----------------------------------------------------------------------------------------------------------|-----------------------------|
| SD = 3.65           | Backed out from dazodalibep’s own Phase 2 standard errors                                                | 0.98 points                 |
| SD = 5.00           | Dazodalibep’s own Phase 2 protocol’s planning assumption                                                 | 1.34 points                 |
| SD = 5.47           | Backed out from ionalumab’s real Phase 3 trial (NEPTUNUS-1) — the only real external benchmark available | 1.47 points                 |

The middle-to-conservative rows are treated as more credible than the first, since the first is the most self-flattering assumption and the other two are either the drug’s own original planning basis or a real outside data point.

### Bull / base / bear (illustrative assumed-effect scenarios, using SD=5.47)

| Scenario    | Assumed true ESSDAI delta | Resulting p-value | Read                                     |
|-------------|---------------------------|-------------------|------------------------------------------|
| <b>Bull</b> | 2.0                       | <0.001            | Would pass clearly                       |
| <b>Base</b> | 1.4                       | ≈0.008            | Would pass, real but not enormous margin |
| <b>Bear</b> | 0.6                       | ≈0.25             | Would not reach significance             |

**These are assumed inputs being stress-tested, not predicted outcomes** — nobody knows the true effect before the readout. The qualitative weighting given to these (Bull ~20% / Base ~50% / Bear ~30%) is an analyst judgment call, explicitly labeled as such, not a computed probability.

### The final call

**We lean positive on NCT06104124's ESSDAI primary endpoint, with material uncertainty that should not be understated.** Ranked reasons: (1) Phase 2's own randomized result is real and appropriately powered for its size; (2) pharmacodynamic biomarker evidence (CXCL13, rheumatoid factor suppression, reversible on crossover) supports genuine target engagement; (3) ianalumab's real Phase 3 wins prove this disease/endpoint combination *can* produce a positive readout, and dazodalibep's own Phase 2 effect already exceeds the size that carried ianalumab through; (4) both CD40-pathway trials to date (dazodalibep, iscalimab) met their respective primary objectives — a real but small-sample pattern, not a computed base rate.

**The single strongest counter-argument:** the only real precedent for calibrating how much a Phase 2 effect shrinks going into Phase 3 (ianalumab, which scaled up  $\sim 1.4\times$ ) retained 68% of its original effect. Dazodalibep's own scale-up is  $\sim 8.8\times$  — far larger, with no precedent for what happens to effect retention at that scale. If erosion is proportionally worse than ianalumab's, the true delta could fall into the exact range (0.6–1.0) where several other Sjögren's trials' real effects have vanished into placebo noise before.

**For NCT06245408** (secondary trial): directionally positive, with more confidence than the primary trial (its own Phase 2 precursor was the cleanest result in the program,  $p=0.0002$ ), a wider error bar (its instruments — ESSPRI and now-added DASPRI — have no objective anchor), and a later, less certain catalyst timeline.

## 4. What tables and charts the presentation will need

Drawn from the reference deck's own visual patterns (see Section 5 below) and this project's actual content. Not yet built — this is the plan for Day 3/4.

| Planned exhibit                                                                    | Data source (already built)                                                                                  |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Trial comparison table (population, N, endpoints, timepoints, status)              | EVIDENCE_TABLE.md, the two trial CSVs                                                                        |
| CD40/CD40L mechanism diagram                                                       | Conceptual — needs to be drawn fresh, no existing file                                                       |
| Phase 2 ESSDAI/ESSPRI longitudinal plots with error bars                           | Data in phase2_results_verified.md; original figures are in the Phase 2 PDF and can be redrawn or referenced |
| Phase 2 domain/responder bar charts                                                | phase2_results_verified.md                                                                                   |
| Ianalumab comparator forest plot/table (Ph2b + NEPTUNUS-1 + NEPTUNUS-2, arm-level) | ianalumab_competitor_analysis.md                                                                             |
| Historical trial comparator matrix                                                 | historical_base_rate_table.md (already qualitative, not a probability table — matches the review's guidance) |
| Variance/power sensitivity table or heatmap                                        | day2_scenario_model.md                                                                                       |
| Bull/base/bear assumption table                                                    | day2_scenario_model.md                                                                                       |
| Phase 2→3 bridge table (what changed, verified vs. assumed)                        | day2_final_thesis.md, phase2_baseline_characteristics.md                                                     |
| Safety summary table (by population/stage, denominators, attribution)              | phase2_results_verified.md                                                                                   |
| Readout interpretation matrix (what would strengthen/weaken/reverse)               | Already fully built in day2_additional_topics.md — ready to drop into a slide with light formatting          |
| Final call slide (headline + ranked evidence + counter-argument)                   | day2_final_thesis.md                                                                                         |

## 5. What the reference deck (MAESTRO-NASH) says, and how we're using it

**File:** MDGL Ph.3 Readout 2022 Sample.pdf (96 pages), a real Jefferies analyst deck by Akash Tewari predicting Madrigal Pharmaceuticals' (MDGL) Phase 3 NASH-drug (resmetirom) readout before it happened.

### What it actually contains

- **A conclusion-first headline on every slide** (e.g., “we think this trial will likely hit on NASH resolution but may be mixed on fibrosis”), never a neutral topic label.
- **A biological-rationale section** connecting the drug's mechanism (a thyroid hormone receptor-beta agonist) to a specific, quantified prediction, not just a disease-education slide.
- **A statistical-design deep dive:** multiple-endpoint testing methods (Bonferroni, Hochberg, Truncated Holm), what the sponsor's own disclosed power assumptions imply about the minimum effect size needed, and how alpha might be split or recycled across two primary endpoints.
- **An explicit Phase 2→Phase 3 bridge table:** starting from Phase 2's actual response rate, then a column-by-column build-up of every design change (updosing, duration, population enrichment, regression to the mean) leading to a Phase 3 response rate — repeated for bull, base, and bear cases with real odds ratios/p-values attached to each.
- **Cross-trial benchmarking:** a table of many other NASH drugs' placebo and drug response rates, used to sanity-check whether the assumed Phase 3 numbers are realistic, not invented.
- **A named “biggest risk” and a dedicated section on what could break the thesis** — not just supporting evidence.
- **A separate valuation section** (investor survey, DCF, scenario-weighted price targets) — explicitly **not** something this project is replicating, per the original scope decision (science/statistics only, no valuation).

### What we are taking as direct reference (methodology, not content)

1. The **conclusion-first slide headline** convention — adopted throughout `day2_final_thesis.md` and planned for every slide in Day 3/4.
2. The **explicit Phase 2→Phase 3 bridge table structure** — directly mirrored in `day2_scenario_model.md`'s bridge-construction table.
3. **Presenting bull/base/bear with real numbers and rationale for each assumption**, not just a qualitative “could go well or badly” — mirrored in the same file.
4. **Cross-trial benchmarking to sanity-check assumptions** — mirrored in `historical_base_rate_table.md` and `ianalumab_competitor_analysis.md`, using real Sjögren's trials instead of NASH trials.
5. A **dedicated, named “strongest counter-argument”** rather than a vague risk list — mirrored in `day2_final_thesis.md`.

### What we've done differently or added beyond the reference deck

1. A **formal variance-sensitivity table across three distinct assumptions**, rather than committing to one. The reference deck's own power/effect-size analysis (pages 26-27) uses a single assumed cohort size and doesn't stress-test the variance assumption itself the way `day2_scenario_model.md` now does.
2. An **explicit statement of what the model does NOT compute** (a true probability of trial success, which would require integrating over effect/variance/dose/missing-data/multiplicity uncertainty) — the reference deck's Monte Carlo section (page 50) does compute an overall probability (83%) for MDGL, which we deliberately did not attempt to replicate for dazodalibep without equivalent disclosed inputs (MDGL's deck had sponsor-disclosed power assumptions to build from; we don't have dazodalibep's SAP).
3. A **live external-review correction cycle** — the reference deck is a finished, polished analyst product; this project's working files carry visible correction notes documenting what was wrong and why, which is

unusual for a finished deck but appropriate for a working research repository.

4. **Pharmacodynamic/biomarker evidence as an independent evidence line** (CXCL13, RF suppression) — the reference deck's biology section supports its *prediction* directly; our project uses biomarker data as a separate check on *whether a future miss should be attributed to the drug or to the endpoint*, which the NASH deck didn't need since MRI-PDFF is itself a quantitative biomarker already built into its prediction chain.
5. **The “clinical meaning of the outcome” synthesis** (day2\_additional\_topics.md, item 2) — distinguishing statistical significance from patient-level benefit from benefit-risk — is a framing this project added explicitly; the reference deck folds an equivalent distinction implicitly into its valuation section instead (which this project is not building).

## 6. Does this satisfy the brief? — Self-review

| Requirement                                                                         | Status            | Detail                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clear call on one or both trials                                                    | ✅ Done            | day2_final_thesis.md — explicit positive lean on NCT06104124, directional positive on NCT06245408<br>Discussed in chat early in the project but <b>never saved to its own file</b> — this is a real gap for a self-contained repo; recommend creating |
| CD40L biological rationale                                                          | 🟡 Partial gap     | biological_rationale.md before Day 3                                                                                                                                                                                                                  |
| Phase 2 analysis                                                                    | ✅ Done            | phase2_results_verified.md, phase2_baseline_characteristics.md, phase2_paper_explained_simple.md                                                                                                                                                      |
| Phase 3 design distinctions (symptomatic vs. systemic) + statistical considerations | ✅ Done            | essdai_measurement_properties.md, nct06245408_esspri_risk_note.md (now includes DASPRI), day2_scenario_model.md                                                                                                                                       |
| Lessons from other trials (ianalumab)                                               | ✅ Done, corrected | ianalumab_competitor_analysis.md, historical_base_rate_table.md                                                                                                                                                                                       |
| Public-source methodology, exact citations                                          | ✅ Done            | Every file carries sources; EVIDENCE_TABLE.md consolidates all of them with uncertainty flags                                                                                                                                                         |
| PowerPoint format                                                                   | ❌ Not started     | Day 3 (outline) and Day 4 (build) remain                                                                                                                                                                                                              |

**Net assessment:** the analytical substance the brief actually asks for is essentially complete and has been independently reviewed and corrected once already. The two real remaining gaps are (1) the missing standalone biological-rationale file, and (2) the entire presentation-building phase, which is a real, substantial remaining effort — not a small finishing touch — given ~3 days left before the deadline.

## 7. Honest risk to the timeline

Day 3 (full slide-by-slide outline) and Day 4 (actual editable .pptx + PDF export) have not started, and building an editable, well-sourced 20-slide deck plus appendix in the remaining time is the largest single piece of work left. Recommend prioritizing getting a complete, even if rough, first-pass deck built soon rather than continuing to deepen the research further.